

NetShield AI

System Architecture & Database Schema

1. System Architecture

NetShield AI follows a layered architecture that separates presentation, backend, business logic, AI inference, data storage, and infrastructure concerns, allowing each layer to scale and evolve independently.

- Presentation Layer – React.js web portal for dashboards, alerts, and analytics
- Backend / API Gateway – FastAPI services handling auth, routing, and access control
- Business Logic Layer – traffic processing, alert orchestration, reporting
- AI / ML Layer – anomaly detection and intrusion prediction models
- Database Layer – PostgreSQL (structured data) + MongoDB (traffic logs, model artifacts)
- Infrastructure Layer – Docker, cloud hosting (AWS/Azure), CI/CD



Figure 1 – NetShield AI High-Level System Architecture

A more detailed, component-level view of this architecture — including the API Gateway, the seven-stage monitoring and AI analytics pipeline, external integrations, data storage, and infrastructure — is provided in Figure 1a.

1a. Detailed Architecture View

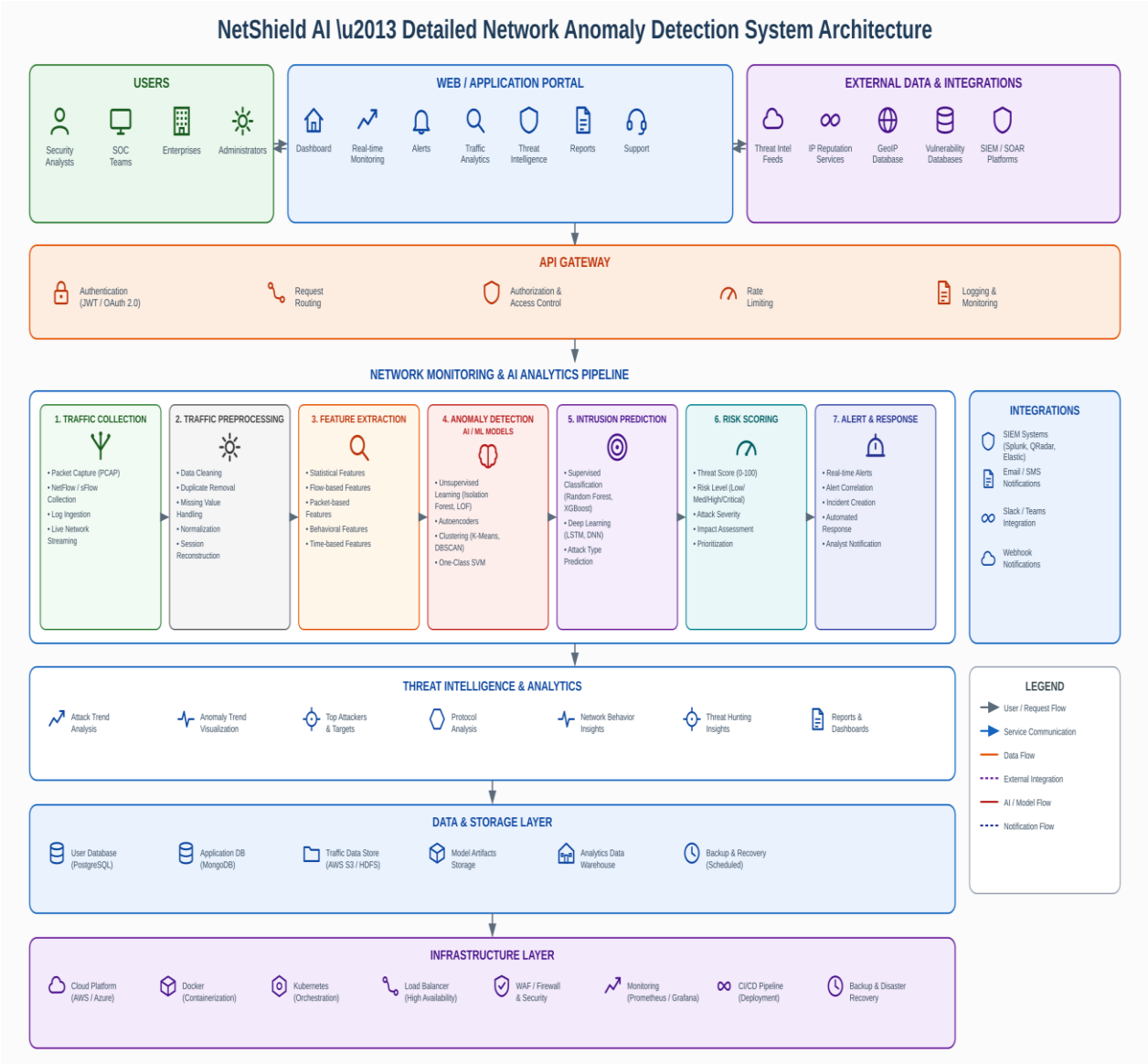


Figure 1a – NetShield AI Detailed System Architecture (component-level view)

2. Database Schema

The conceptual schema below defines the core entities and relationships. It is design-level only; no SQL is included.

Entity	Description	Key Relationships
User	Platform account (analyst, manager, admin)	N:1 Role; 1:N Audit Log, Alert, Incident
Role	Named permission set	1:N User
Network Traffic Log	Processed traffic/flow records	1:N Anomaly Event
Anomaly Event	Detected deviation with confidence score	N:1 Traffic Log, Model; 1:1 Alert
Alert	Actionable, prioritized notification	1:1 Anomaly Event, Incident; N:1 User
Incident	Logged resolution of an alert	1:1 Alert; N:1 User
Detection Model	Trained AI model version	1:N Anomaly Event
Audit Log	System/user action record	N:1 User
System Configuration	Platform config key/value store	Standalone
Threat Intelligence Feed	(Future) external enrichment indicators	N:N Anomaly Event

NetShield AI - Conceptual Entity-Relationship Diagram

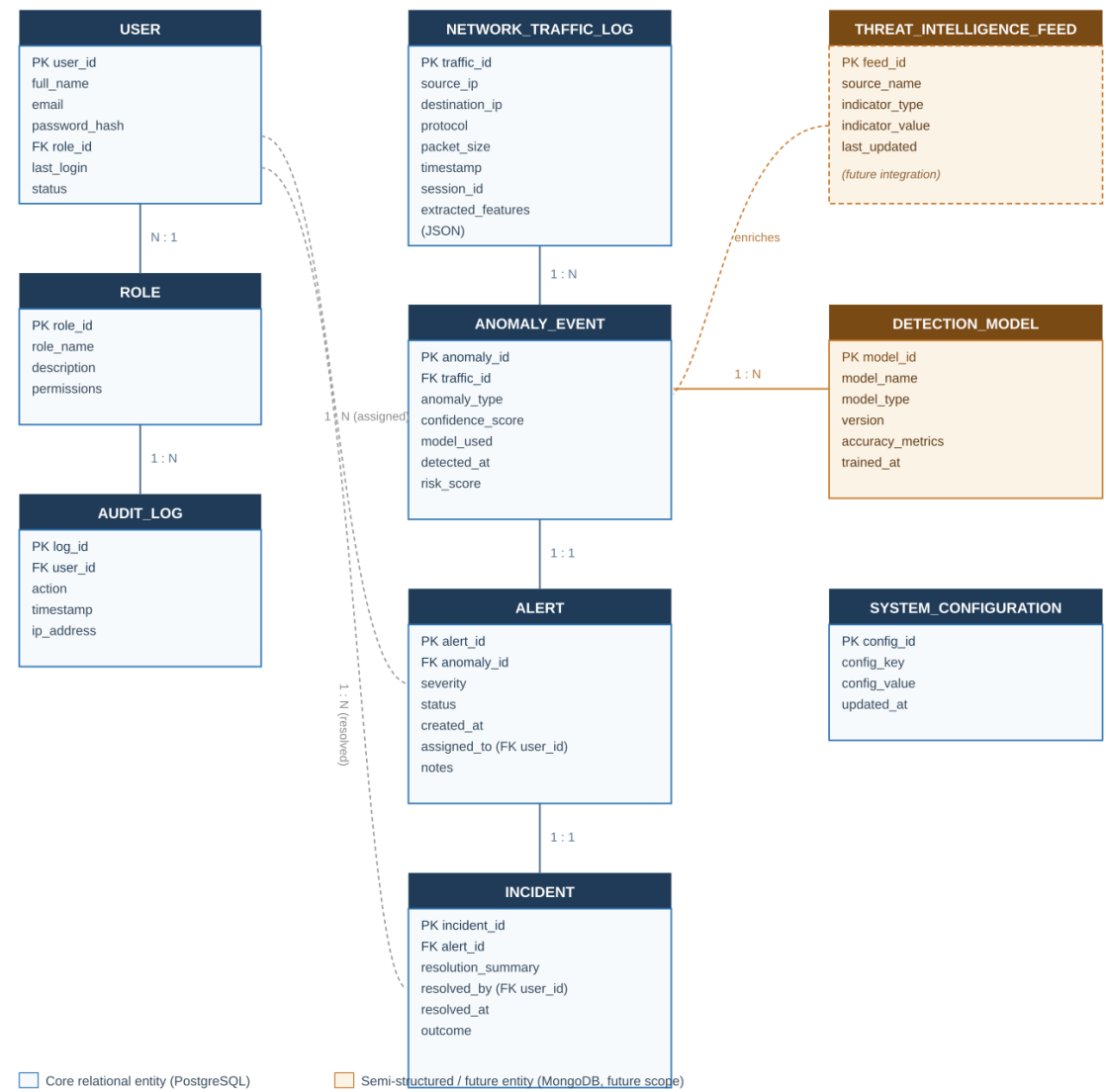


Figure 2 – NetShield AI Conceptual Entity-Relationship (ER) Diagram